

# **STEAM** Sharing: M5StickS3

Xiaozhi · UIFlow 2 · Vibe Coding



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YL Long Ping Estate Wai Chow School

22 Jun 2026 · 14:00–15:30

Presenter: Warren Chan · Carmel Holy Word Secondary School

# 立即加入 Google Classroom



課程內容：Task 3 Sample Prompt · 跟做筆記 · Feedback Form · 有用連結

- ② 按「+」→「以代碼加入課程」
- ③ 輸入代碼：zgabqpbr
- ④ 或直接掃描左邊 QR Code

**Class Code : zgabqpbr**

# About Me

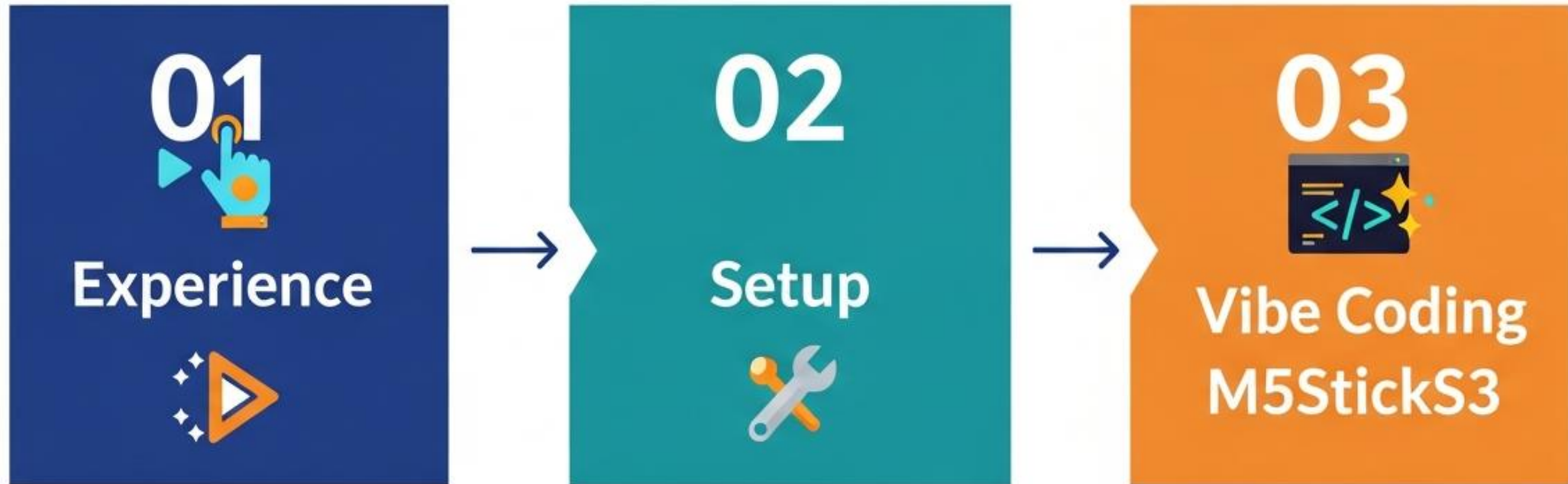
## Warren Chan

Carmel Holy Word Secondary School

- Software Engineer — 10+ years
- Teaching Secondary School — 9+ years
- Work for NGO and startups
- Mentored well-known university students — MIT, Stanford, Chicago, and more



# Three Tasks



# Know These Three Buttons



## Power Button

Single press to power ON

Double press to sleep

Hold 6 seconds to download mode (DON'T DO THAT)

## Button A

Blue button (blue)

Normally means enter/select

## Button B

Right side button

Single press means next.

Double press means previous.

# M5StickS3 – More Than a Toy

 **Color Screen**

 **ESP32-S3 WiFi**

 **Microphone**

 **Speaker**



 **6-axis IMU**

 **IR transmitter**

 **RTC**

 **USB-C battery** 

Full IOT **computer** in your palm



# Hands-on Setup



20 devices only ~ M5Stack stock limited



**Please form groups  
of 3 teachers**

# Task 1: Feel the Power

01



Press Power

02



Wait

03



Talk

Xiaozhi = English tutor

5 min  
explore



# Try Saying...



**Turn on  
dark mode**



**Play a  
song**



**What's the  
weather?**



**Teach me  
English**

# Notice

Conversations may be recodred for teaching quality

Use brief test phrases if concerend

# # Why So Powerful?



Voice



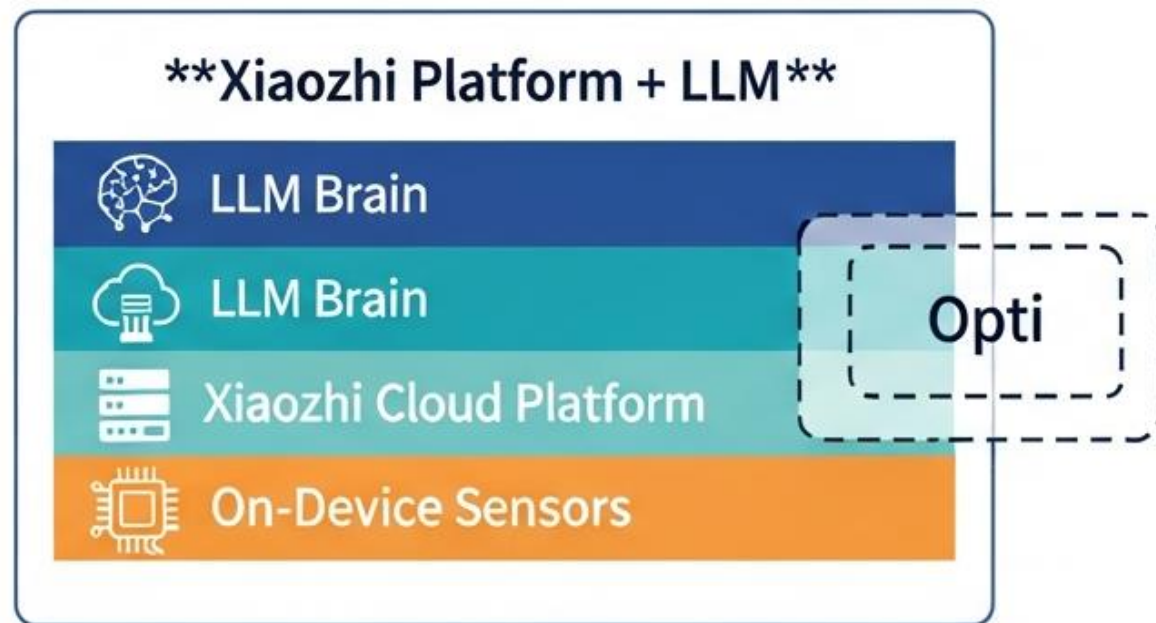
WiFi



Microphone



WiFi



xiaozhi.me

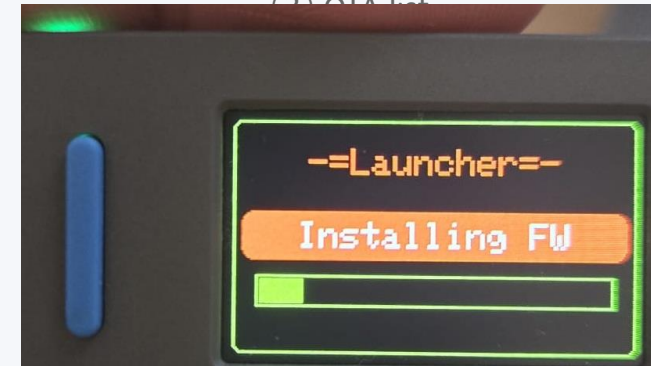
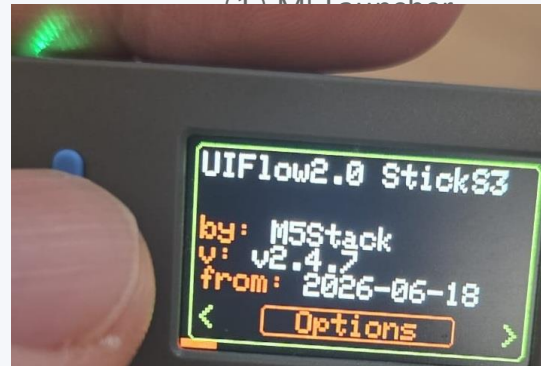
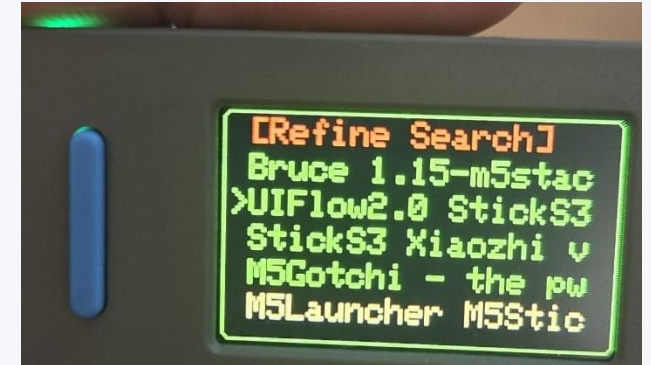
## Show the Xiaozhi device management dashboard

[illegible]

## UIFlow 2 IDE — used in Task 2 & 3

# Task 2: Install UIFlow 2

- ① Power ON → M5Launcher appears
- ② Press Button B immediately
- ③ Select OTA in menu
- ④ Choose UIFlow 2 from list
- ⑤ Wait for install to complete



⌚ Takes about 1–2 minutes



# Imagine...

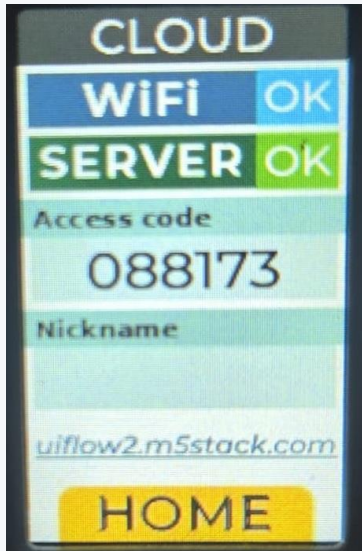


## Endless STEAM projects



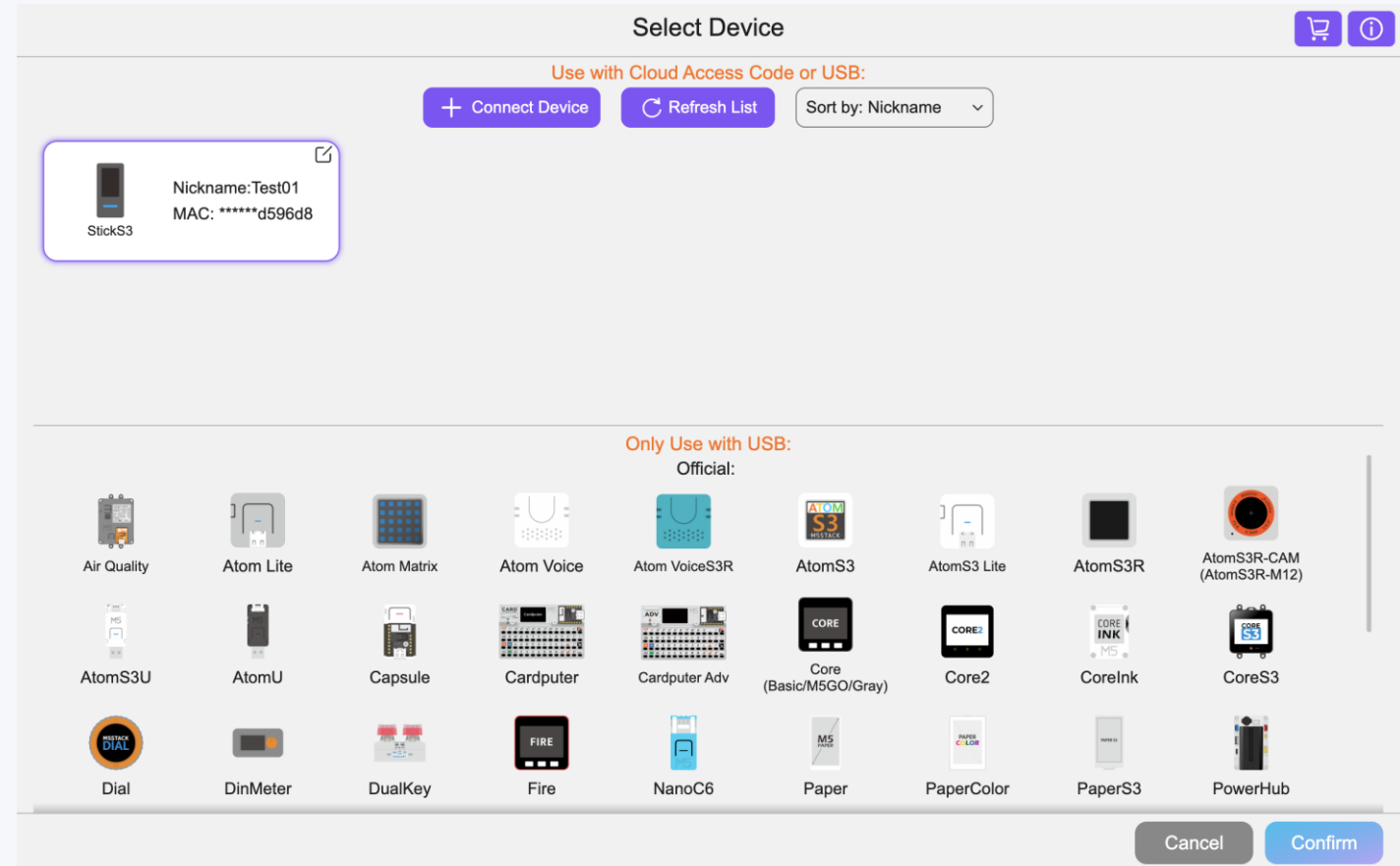
# Pair with UIFlow 2

- ① On device: press Cloud icon
- ② Note the 6-digit Access Code
- ③ Open [uiflow2.m5stack.com](https://uiflow2.m5stack.com)
- ④ Click Connect Device → enter code
- ⑤ Click Run Once to test



② Device shows Access Code

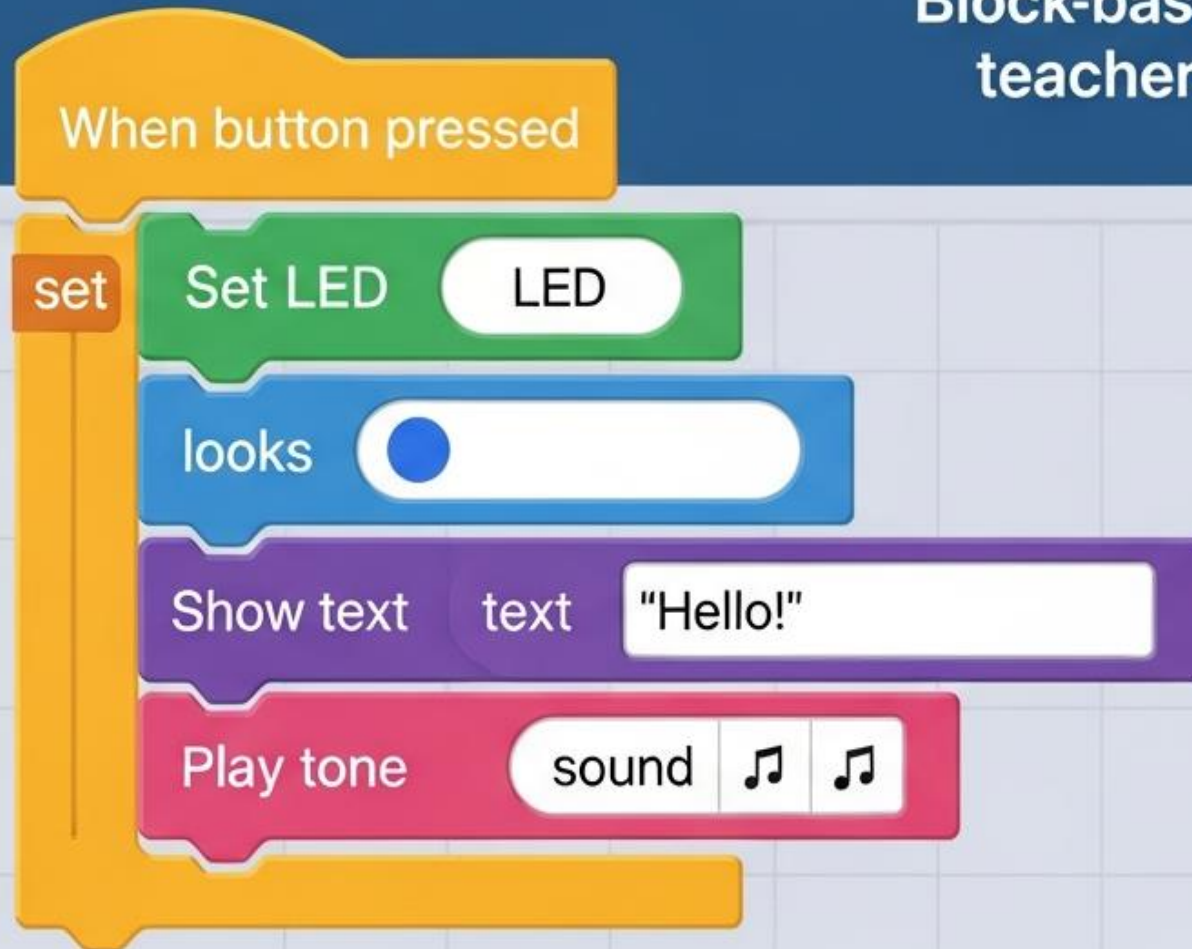
[uiflow2.m5stack.com](https://uiflow2.m5stack.com)



④ UIFlow 2: Connect Device

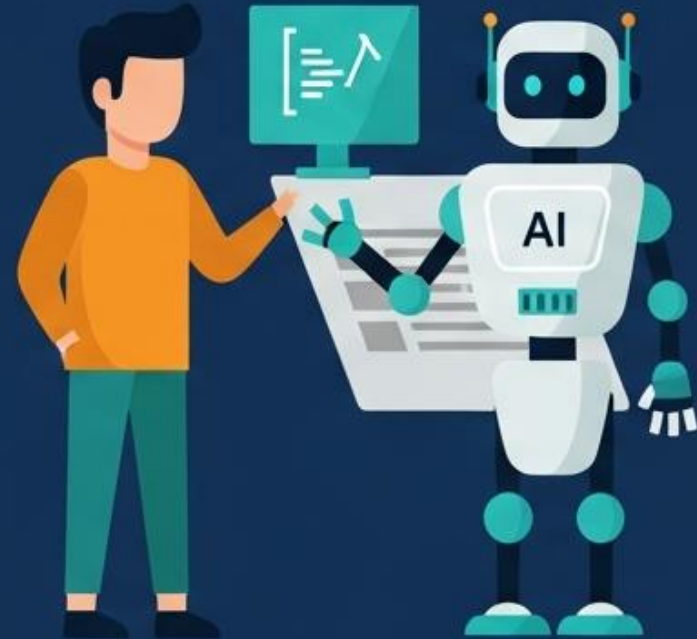
# Sound Familiar?

Block-based web IDE — many STEAM teachers already know this



# Honest Truth from a Software Engineer

- “ Industry rarely hand-writes code anymore
- “ We review AI output — don't break the codebase



# The Evolution of AI Interaction



**Chatbot**  
**"Use"**

Xiaozhe:  
Example



**Vibe Coding**  
**"Create"**

Gemini game  
Example

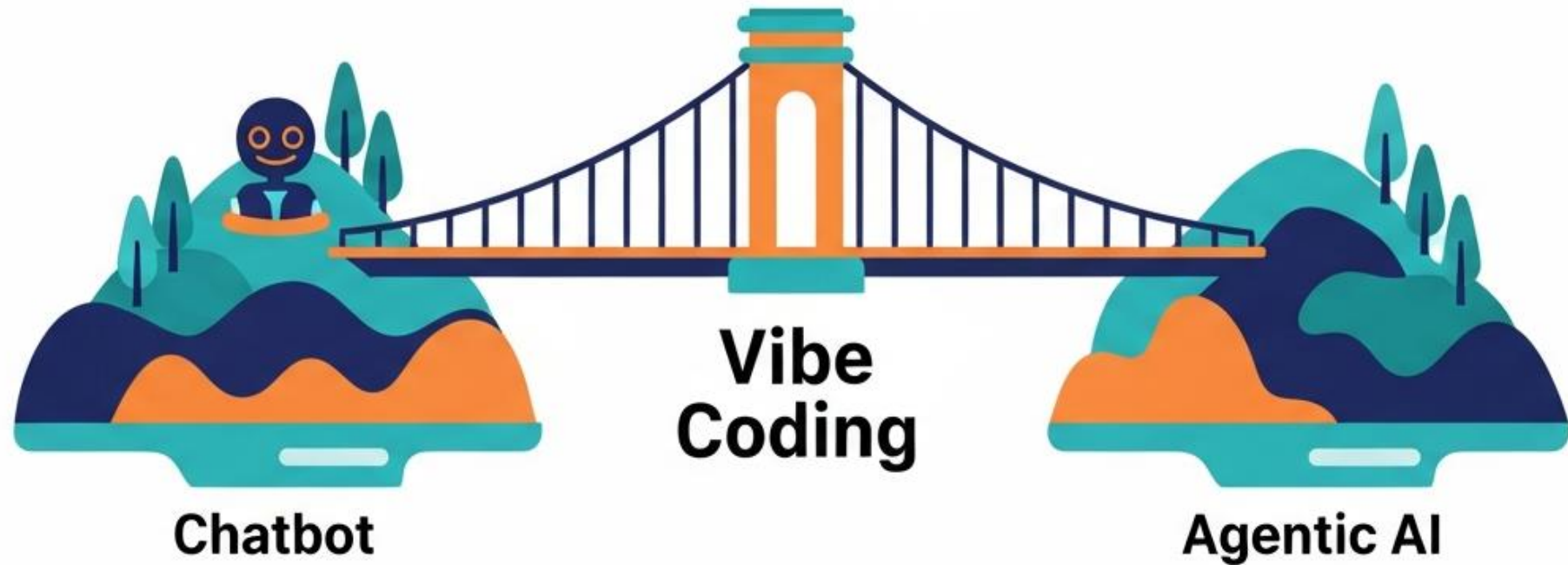


**Agentic AI**  
**"Automate"**

Example:  
Multi-step agent

# Vibe Coding = The Bridge

Experience before you can teach AI what to do





# Task 3: Vibe Coding



Sample prompts posted

1.

Copy

2.

Paste

3.

Generate





## Pro Tip

請參考 M5StickS3 的官方 UIFlow2 MicroPython API 文件：

<https://uiflow-micropython.readthedocs.io/en/latest/controllers/sticks3.html>

我需要製作一個「考反應的遊戲」

# Design Constraint



Limited gamepad input:  
Only BtnA + BtnB

Versus



Add voice < new possibilities

# DeepSeek API – Next Session



- DIY backend
- Slower than Xiaozhi
- Great for understanding APIs

# Want to Go Deeper?

My next workshop: Agentic AI for Educators

## ① Chatbot

今日已做 ✓

小智

Ready-made AI  
Use it in class



## ② Vibe Coding

今日已做 ✓

Gemini + UIFlow

Describe → AI codes  
You review & run



## ③ Agentic AI

下一步 →

Claude · Cursor

Give a goal →  
AI plans & acts

業界已唔人手逐行寫code — 你定目標, AI 計劃並執行。有興趣? 歡迎聯絡 Warren Chan。

# # Takeaways

- 1  **Xiaozhi** = ready-made AI = Icon 1
- 2  **UIFlow 2** = student creativity
- 3  **Vibe Coding** = how we build now



M3StickS3



# Questions?

# Thank you!

STEAM Sharing: M5StickS3



YL Long Ping Estate Wai Chow School  
Warren Chan  
Carmel Holy Word Secondary School

# 課後 Feedback — 請掃 QR Code

STEAM 分享：M5StickS3 · 2026 年 6 月 22 日 · 約 3–5 分鐘



**用手機掃描左邊 QR Code**

填寫工作坊 feedback form

或睇 Google Classroom

搵「課後 Feedback」連結

**多謝各位同事！**

有問題歡迎聯絡 Warren Chan

[https://docs.google.com/forms/d/e/1FAIpQLSff43JQ-OeTZiGcfMKEM-Pyq\\_ByyzZ88Tekn4Y1kgGgcDYihw/viewform](https://docs.google.com/forms/d/e/1FAIpQLSff43JQ-OeTZiGcfMKEM-Pyq_ByyzZ88Tekn4Y1kgGgcDYihw/viewform)